Confidence Calibration Test Classroom Run Kit

TOK Cognitive Science Activity Bank • Knowledge and the Knower • 20-25 min

Category	Knowledge and the Knower	**Time**	20-25 min
Difficulty	Easy	**ID**	confidence-calibration-test

Big idea: Certainty and accuracy are not the same.

One-Minute Teacher Brief

Students compare how confident they feel with how accurate they actually are.

Student Prompt

Answer each question, then mark your confidence: 50%, 60%, 70%, 80%, 90%, or 100%.

Concrete Stimulus Packet

Student Prompt	Answer each question privately. After marking, calculate accuracy within each confidence band.	Do not allow lookup; the point is calibration, not trivia mastery.
Question 1	Which planet is closest to the Sun?	Confidence: 50% / 60% / 70% / 80% / 90% / 100%
Question 2	What is the capital city of Canada?	Confidence: 50% / 60% / 70% / 80% / 90% / 100%
Question 3	Which is heavier: a kilogram of feathers or a kilogram of steel?	Confidence: 50% / 60% / 70% / 80% / 90% / 100%
Question 4	In which year did the Berlin Wall fall?	Confidence: 50% / 60% / 70% / 80% / 90% / 100%
Question 5	What is the chemical symbol for potassium?	Confidence: 50% / 60% / 70% / 80% / 90% / 100%
Question 6	Which country has the longest coastline in the world?	Confidence: 50% / 60% / 70% / 80% / 90% / 100%
Question 7	How many bones are in the adult human body?	Confidence: 50% / 60% / 70% / 80% / 90% / 100%
Question 8	What is the only mammal capable of true sustained flight?	Confidence: 50% / 60% / 70% / 80% / 90% / 100%
Question 9	Which has more letters: the Greek alphabet or the modern English alphabet?	Confidence: 50% / 60% / 70% / 80% / 90% / 100%
Question 10	Without looking it up, what is the 19th digit after the decimal point in pi?	Confidence: 50% / 60% / 70% / 80% / 90% / 100%

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	Which planet is closest to the Sun?	Mercury
2	What is the capital city of Canada?	Ottawa
3	Which is heavier: a kilogram of feathers or a kilogram of steel?	Neither — they weigh the same
4	In which year did the Berlin Wall fall?	1989
5	What is the chemical symbol for potassium?	K
6	Which country has the longest coastline in the world?	Canada
7	How many bones are in the adult human body?	206
8	What is the only mammal capable of true sustained flight?	Bat
9	Which has more letters: the Greek alphabet or the modern English alphabet?	English alphabet — 26 vs Greek 24
10	Without looking it up, what is the 19th digit after the decimal point in pi?	4

Student Task

Complete the quiz, mark confidence for every answer, then compare confidence bands with actual accuracy.

Teacher Key / Reveal

Where were we most confident but least accurate, and what does that reveal about certainty as a way of knowing?

• Q1: Mercury

• Q2: Ottawa

• Q3: Neither — they weigh the same

• Q4: 1989

• Q5: K

• Q6: Canada

• Q7: 206

• Q8: Bat

• Q9: English alphabet — 26 vs Greek 24

• Q10: 4

Student Instructions

• Work silently for the first response so you can notice your own starting point.

• Record one knowledge claim, the evidence behind it, and your confidence level.

• After the reveal or comparison, update your claim in a different colour or column.

• Use the debrief to answer: Is certainty a reliable indicator of knowledge?

Setup Checklist

• Prepare a starter stimulus using: Ten general knowledge questions with confidence bands from 50% to 100%.

• Print or project the student response grid before the activity begins.

• Plan a private first-response moment before any group discussion.

Example Stimuli or Materials

• Ten general knowledge questions with confidence bands from 50% to 100%.

• A mini quiz mixing easy, medium, and impossible questions so overconfidence becomes visible.

• A class calibration chart comparing confidence level with actual accuracy.

Resources To Use

• Resource pack: resources/confidence-calibration-test/index.html

• Card deck CSV: resources/confidence-calibration-test/cards.csv

• Visual prompt: resources/confidence-calibration-test/visual-prompt.svg

• Printable student worksheet with observation, inference, evidence, and confidence columns.

• Teacher notes with setup checklist, sample facilitation language, misconceptions, and assessment look-fors.

• Classroom run kit with prompt cards, example stimuli, and debrief moves.

• PowerPoint deck with a student-facing sequence for running the activity.

Run Sequence

1. Give students general knowledge questions. For each answer they also write a confidence level: 50%, 60%, 70%, 80%, 90% or 100%.

2. Mark the answers.

3. For each confidence band, calculate actual percentage correct.

4. Ask students whether they were overconfident, underconfident or well calibrated.

Debrief Moves

• Knowledge question: Is certainty a reliable indicator of knowledge?

• What is the difference between confidence, belief and justification?

• Why might experts sometimes be better calibrated than beginners?

Teacher Quick Script

• Write your first judgement before we discuss it; the first answer is useful evidence.

• Separate what you noticed from what you inferred.

• What would make this knowledge claim more reliable?

Exit Ticket

• One thing I first treated as evidence was...

• My confidence changed because...

• A better knowledge question I can now ask is...